1. Introduction

Women's safety in public spaces is an important social concern, as incidents of harassment may occur repeatedly in particular locations. Identifying such locations from available incident records can help authorities take preventive measures and improve patrolling. However, when incident information is collected continuously, manually analysing the records and identifying important locations becomes difficult.

The proposed project, “Incident Analysis and Hotspot Identification for Security Provision Against Women Harassment in Public Spaces,” aims to develop a data-driven system for analysing reported incidents and identifying areas where incidents are concentrated or repeatedly reported.

This synopsis focuses on Module 1 – Incident Analysis, Geospatial Analysis and Hotspot Identification, which represents the analytical part of the overall system. The module uses an existing Excel sheet containing geographical and public infrastructure information based on OpenStreetMap (OSM), along with assumed/simulated safety-infrastructure attributes such as lighting and surveillance. These infrastructure attributes are prototype/demo data and are not treated as verified real-world infrastructure information.

New incident information submitted through the safety mobile application is also integrated into the system. The module analyses incident history based on location, time, frequency and recurrence. Geospatial analysis is performed to understand the spatial distribution of incidents and identify concentrated locations. The results are used for risk-level assessment and are provided through APIs for visualization in the authority dashboard.

The proposed system focuses on historical incident analysis and hotspot identification rather than crime prediction. The main purpose is to provide evidence-based information that can help authorities concentrate preventive patrolling and safety measures in locations where incidents have been repeatedly or highly concentrated.

2. Literature Survey

2.1 Geospatial Analysis for Public Safety

Geospatial analysis is widely used to understand how incidents are distributed across geographical locations. Mapping incident coordinates helps identify areas where incidents are concentrated and allows the surrounding geographical environment to be studied.

Geographical information can provide additional context about an incident location, including nearby roads, streets, buildings and other public features.

2.2 Hotspot Identification

Hotspot identification techniques are used to locate areas having a high concentration of reported incidents. Such analysis can assist authorities in prioritising locations for monitoring and preventive action.

Hotspots can be identified by analysing the spatial concentration and recurrence of reported incidents. This approach helps understand historical incident patterns without attempting to predict future incidents.

2.3 Incident Data Analysis

Incident data analysis involves examining records based on parameters such as incident type, date, time, location and frequency. Analysing historical incident records can reveal recurring patterns and help identify locations that require additional attention.

Structured analysis also allows authorities to compare incident activity across different geographical areas.

2.4 Geographic Information and OpenStreetMap

Geographical information provides important context for incident analysis. OpenStreetMap provides geographical features such as roads, streets, buildings and other public features that can be used to understand the surroundings of reported incidents.

Integrating geographical information with incident records can improve location-based analysis and visualization.

2.5 Mapping and Visualization

Maps and graphical visualizations provide an effective way to communicate analytical results. Location-based incident information can be represented using interactive maps, charts and statistics so that authorities can quickly understand affected areas.

Map-based visualization is particularly useful for representing incident concentrations and hotspots.

2.6 Continuous Data Updating

A safety reporting system may receive new incident records continuously. Therefore, an analytical system should be capable of incorporating newly reported incidents and updating relevant statistics and hotspot information.

Continuous updating makes the system more useful for monitoring incident activity over time.

2.7 Research Gap

Existing systems often focus on incident reporting or basic visualization separately. There is a need for an integrated analytical approach that combines incident records, geographical information, infrastructure-related attributes and continuously updated incident reports to support historical analysis, hotspot identification and preventive patrolling.

3. Problem Statement, Aim and Objectives

3.1 Problem Statement

Women-safety incidents reported in public spaces may occur repeatedly or become concentrated in specific locations. Without systematic analysis of incident records and geographical information, authorities may find it difficult to identify such locations and prioritise preventive patrolling.

As incident reports continue to increase, manually analysing the records becomes time-consuming and makes it difficult to identify recurring patterns and geographical concentrations.

Therefore, there is a need for an analytical module that integrates incident and geographical data, analyses historical incident patterns, identifies concentrated and recurring locations, assesses their risk level and continuously updates the results when new incidents are reported.

3.2 Aim

To develop an analytical module that integrates incident and geographical data, performs incident and geospatial analysis, identifies concentrated and recurring women-safety hotspots, and provides updated location-based information to support preventive patrolling.

3.3 Objectives

- To integrate the existing Excel-based geographical and safety-infrastructure data, including OSM-based features and assumed/simulated lighting and surveillance information.
- To collect and integrate new women-safety incident reports submitted through the mobile application.
- To preprocess the integrated data by handling missing, duplicate and inconsistent records.
- To analyse reported incidents based on location, time, frequency and recurrence.
- To perform geospatial analysis to understand the spatial distribution of reported incidents.
- To identify locations where incidents are concentrated or repeatedly reported and classify their risk level.
- To automatically incorporate newly reported incidents into the analysis and update hotspot and statistical information.

- To provide updated analytical results through APIs for visualization in the authority dashboard.
- To design the analytical module so that it can be extended to different geographical regions and cities.

ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ
BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
(Autonomous Institution Under VTU)
Yelahanka, Bengaluru -560119

Department of MCA
(Accredited by NBA, New Delhi)

4. SDG & ESG Compliance

A. Section A — SDG Compliance

i. Project title: Incident Analysis and Hotspot Identification for Security Provision Against Women Harassment in Public Spaces

ii. Identified SDG (Goal No. & Name):

Sustainable Development Goal (SDG): SDG 5 – Gender Equality

iii. Target & Indicator under this SDG:

Target 5.2 – Eliminate all forms of violence against all women and girls in public and private spheres, including trafficking, sexual exploitation, and other forms of violence.

Indicator 5.2.2 – Proportion of women and girls subjected to sexual violence by persons other than an intimate partner, by age and place of occurrence.

iv. Contribution of the Project towards the identified SDG:

This project aligns with SDG 5 by addressing women's safety in public spaces and supporting preventive measures against harassment. Its contribution can be summarised as follows:

- The system analyses women-safety incidents and identifies locations where incidents are concentrated or repeatedly occurring.
- Hotspot visualization and automated alerts give authorities information that can support preventive patrolling and timely intervention.

- The mobile application delivers safety-oriented functionality to users, contributing to safer participation in public spaces.

v. Supporting Documents / Evidence:

- Project synopsis describing the system's objectives, methodology and alignment with SDG 5 - Gender Equality.
- Incident dataset and sample incident records used to analyse women-safety incidents in public spaces.
- Geospatial maps and hotspot visualizations showing locations with concentrated or recurring incidents.
- Statistical charts and analysis results demonstrating incident frequency, recurrence and geographical distribution.
- Screenshots of the authority dashboard displaying incident statistics, hotspot locations and risk-level information.
- Screenshots of the mobile safety application demonstrating safety-oriented features for users.
- Sample automated alerts generated from incident history to highlight areas requiring increased attention.
- Sample outputs demonstrating how the analytical results can support preventive patrolling and safety planning.

ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ
BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
(Autonomous Institution Under VTU)
Yelahanka, Bengaluru -560119

Department of MCA
(Accredited by NBA, New Delhi)

B. Section B — ESG Compliance

ESG Category	Project Contribution	Supporting Documents
Environmental (E)	The system uses digital reporting and storage instead of depending on paper records. This helps reduce paper usage and makes incident records easier to maintain digitally.	System design, database design, web/mobile application screenshots
Social (S)	The system helps authorities and users understand where and when women's safety incidents occur more frequently. By identifying recurring incidents and high-risk areas, it supports better patrolling and safety planning. Interactive maps and dashboards make the information easier to understand and help authorities make more informed decisions based on actual incident data.	Project synopsis, hotspot maps, authority dashboard, mobile application screenshots, SDG alignment
Governance (G)	The system protects incident information by using authentication and role-based access. This ensures that only authorized users can access the data and helps maintain the privacy and security of sensitive information.	System architecture, authentication module, access-control design, security documentation

Signature of Student **Signature of Project Guide**

5. Methodology

The proposed Module 1 follows a continuous data-processing and analysis workflow:

Existing Excel Data + Mobile App Incident Data → Backend/API → Database → Data Pre-processing → Incident Analysis → Geospatial Analysis → Hotspot Identification → Risk-Level Assessment → Automatic Analysis Update → API & Result Delivery → Authority Dashboard

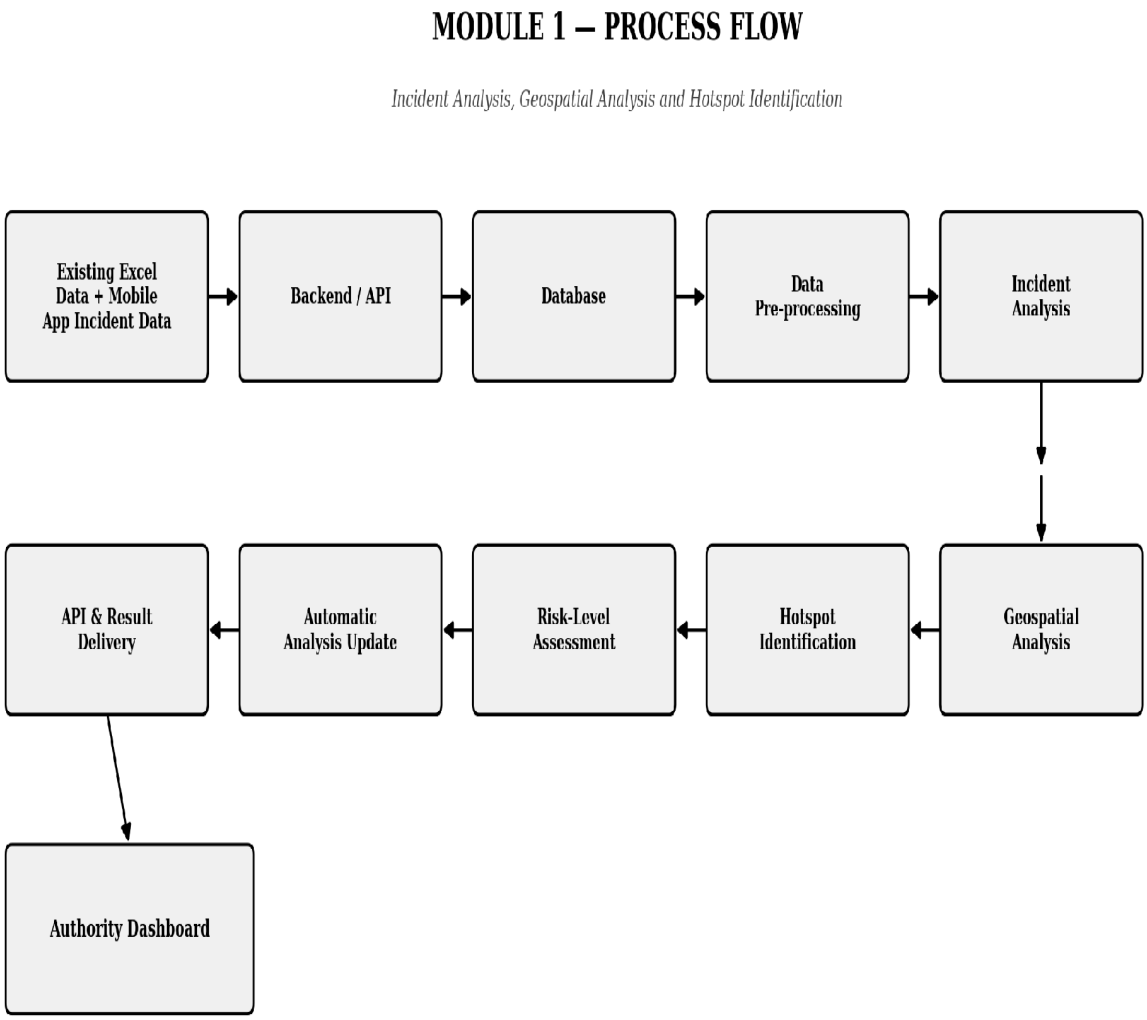


Fig. 1: Module 1 process flow, from data integration to authority dashboard delivery

5.1 Data Integration

The existing Excel sheet contains geographical and public infrastructure information based on OpenStreetMap, such as roads, streets and buildings. It also contains assumed/simulated safety-infrastructure attributes such as lighting and surveillance.

New incident reports submitted through the mobile application are received through the backend/API and integrated with the existing data. Thus:

Existing Excel Data + New Incident Data → Integrated Dataset

This provides a common dataset for subsequent analysis.

5.2 Data Pre-processing

The integrated data is cleaned and prepared before analysis. The preprocessing stage includes handling missing values, removing duplicate records, standardising formats, validating latitude and longitude, maintaining consistent incident categories and checking data consistency.

This ensures that the data is structured and suitable for analytical and geospatial processing.

5.3 Incident Analysis

The incident analysis stage examines historical incident records using parameters such as incident type, date, time, location, frequency and recurrence.

For example, if one street has 12 reported incidents while another has only 2, the first location shows higher historical incident activity.

This stage helps identify locations with frequent or repeated incidents and does not involve future crime prediction.

5.4 Geospatial Analysis

The geographical coordinates of incidents are used to analyse their spatial distribution.

Latitude and longitude values allow reported incidents to be mapped and studied along with surrounding geographical features obtained from OpenStreetMap, such as roads, streets and buildings.

This helps understand the relationship between incident locations and their surrounding geographical environment.

5.5 Hotspot Identification

The system analyses the spatial concentration and recurrence of historical incidents to identify hotspots.

Locations having a higher concentration or repeated occurrence of incidents can be highlighted on maps and through statistical results.

The hotspot identification is based on historical reported incidents and does not predict future incidents.

5.6 Risk-Level Assessment

The risk level of different locations is assessed using factors such as incident frequency, recurrence and spatial concentration.

Locations can be categorised into:

Low Risk → Medium Risk → High Risk

This classification helps authorities prioritise areas requiring greater preventive attention.

5.7 Automatic Analysis Update

When a new incident is reported through the mobile application, it is sent to the backend through an API and stored in the database. The new record is then incorporated into the analytical process. The continuous flow is:

Mobile App → API → Backend → Database → Analysis Update → Hotspot/Statistics Update

This allows the system to maintain updated incident statistics, hotspot information and risk-level information.

5.8 API & Result Delivery

The processed analytical results are exposed through REST APIs using FastAPI. The APIs can provide:

- Incident statistics.
- Incident locations.
- Hotspot locations.
- Risk levels.
- Geographical information.

The authority dashboard retrieves these results and displays them through maps, charts and statistics.

5.9 Region-Based Analysis

The system supports analysis based on selected geographical regions.

For example, an authority can select a particular region such as South Bangalore or North Bangalore. The system can then display the corresponding incident statistics, hotspots and risk-level information for the selected area.

This allows authorities to focus their analysis on specific regions.

5.10 Preventive Patrolling Support

The analytical results can support preventive patrolling by highlighting locations with repeated or concentrated historical incidents.

For example, if a particular area has consistently recorded several incidents, authorities can give greater attention to that area and consider increased preventive patrolling.

Thus, the system supports evidence-based preventive action rather than future crime prediction.

6. Modules Identified

The following modules are identified under Module 1 – Incident Analysis, Geospatial Analysis and Hotspot Identification. Each module performs a specific function in the analytical workflow.

6.1 Data Integration

This module combines the existing Excel-based geographical and infrastructure data with new incident reports received from the mobile application. The existing data includes OpenStreetMap-based geographical information and assumed/simulated safety-infrastructure attributes such as lighting and surveillance. The integrated data provides the common input required for subsequent processing and analysis.

6.2 Data Pre-processing

This module prepares the integrated data for analysis. It handles missing values, duplicate records, inconsistent formats and invalid geographical coordinates. The module ensures that the data is clean, consistent and properly structured before performing incident and geospatial analysis.

6.3 Incident Analysis

This module analyses historical incident records based on parameters such as incident type, location, date, time, frequency and recurrence. The analysis helps determine where and when incidents have occurred and identifies locations with higher or repeated incident activity.

6.4 Geospatial Analysis

This module analyses the geographical distribution of reported incidents using their latitude and longitude coordinates. The incident locations are studied along with surrounding geographical features obtained from OpenStreetMap, such as roads, streets and buildings, to understand the spatial characteristics of incident locations.

6.5 Hotspot Identification

This module identifies locations where incidents are spatially concentrated or repeatedly reported. The identified hotspots are based on historical incident records and can be represented using maps

and statistical information. The module is intended to identify existing patterns rather than predict future incidents.

6.6 Risk-Level Assessment

This module assesses the relative level of attention required for different locations based on factors such as incident frequency, recurrence and concentration. Based on these factors, locations can be classified into categories such as Low Risk, Medium Risk and High Risk. This helps authorities prioritise areas for preventive safety measures.

6.7 Automatic Analysis Update

This module ensures that new incident reports are automatically incorporated into the existing analytical results. Whenever a new incident is submitted through the mobile application, the information is transferred to the backend, stored in the database and included in the analysis. Relevant incident statistics, hotspot information and risk-level information can then be updated.

6.8 API & Result Delivery

This module delivers the processed analytical results to the authority dashboard through APIs. The module provides information such as incident statistics, incident locations, identified hotspots, risk levels and geographical information. The dashboard can retrieve these results and present them through maps, charts and other visualizations.

Together, these eight modules provide a complete analytical workflow from data integration and preprocessing to incident analysis, hotspot identification, continuous updating and result delivery.

7. Tools and Technologies Identified

Table No: 7.1 Tools and technologies

Tool / Technology	**Purpose**
Python	Data processing and analytical implementation
FastAPI	Backend and REST API development
MongoDB	Storage of incident and processed data
Pandas	Data manipulation and preprocessing
NumPy	Numerical processing
GeoPandas	Geospatial data processing
Folium / Leaflet	Interactive map visualization
Matplotlib / Plotly	Charts and analytical visualization
OpenStreetMap	Geographical and public infrastructure information
Microsoft Excel	Existing geographical and infrastructure dataset
JSON / CSV	Data exchange and storage formats
Postman	API testing
Git & GitHub	Version control and collaboration
Visual Studio Code	Development environment

8. Timeline using Gantt Chart

Table No: 8.1 Gantt Chart

Activity	W1-W2	W3-W4	W5-W6	W7-W8	W9-W10	W11-W12	W13-W14	W15-W16
Problem Identification & Requirement Analysis	✓							
Literature Survey	✓	✓						
Existing Excel Data Study		✓						
Data Integration		✓	✓					
Data Pre-processing		✓	✓					
Incident Analysis			✓	✓	✓			
Geospatial Analysis				✓	✓			
Hotspot Identification				✓	✓	✓		
Risk-Level Assessment					✓	✓		
Automatic Analysis Update						✓	✓	
API & Result Delivery						✓	✓	
Testing & Validation							✓	✓
Documentation & Demonstration								✓

Expected Deliverable

At the end of the implementation, Module 1 is expected to deliver a functional analytical component for integrating geographical, infrastructure and incident data and performing systematic incident and geospatial analysis.

The module will provide cleaned and structured incident data, historical incident statistics, geographical analysis, hotspot identification and risk-level assessment. It will identify locations where incidents are concentrated or repeatedly reported and provide location-based information to support preventive patrolling and safety planning.

The module will also support region-based analysis, allowing authorities to select and analyse different geographical regions or cities based on the available geographical and incident data. The current implementation will use the available Bengaluru dataset along with incident data collected

through the mobile application, while the analytical workflow will remain applicable to other regions and cities.

Whenever new incident reports are received through the mobile application, the module will automatically incorporate the new records and update the relevant incident statistics, hotspot information and risk levels. The processed results will be exposed through APIs for visualization in the authority dashboard.